Today: redundancy(!) and what to do about it

<u>Functional dependency</u>: one attribute's value depends on another.

→ Yes!

An example:

Student	Year	Address	Class Dean
Gideon	26	—	Carlson
Luke	27	—	Carlson
Oseas	28	—	Fleming
Bryce	26	—	Carlson

26 → Carlson

28 → Fleming

Year → Class Dean

Both columns are redundant
storing same info, over and over

A FD $X \to Y$ holds for a reln R if
for all allowable instances of that
reln, and for all tuples t_1, t_2 in R

If $\Pi_X(t_1) = \Pi_X(t_2) \Rightarrow$

$$\Pi_Y(t_1) = \Pi_Y(t_2)$$

A FD is baked into the design of a
reln. If you require it to be there, it's
true for all possible instances.

You can't guess it's there just by looking
at some data.

A common fix when you have redundant data due to a FD is to decompose the reln into multiple relations.

An example:

student	year	address	class dean

⋮

⇓ decompose into

student	year	address
gideon	26	—
luke	27	—
bryce	26	—
oseas	28	—

year	class dean
26	carlson
27	carlson
28	fleming

Goal:- Define what it means for a reln to be "redundancy good" vs "redundancy bad"

- Understand how to fix, and what the issues are.

Codd defined (others followed) a variety of <u>normal forms</u>, which are ways of describing good relns.

How do we get thee?

We need to be able to express all of the FDs that exist in a reln.

$R(X, Y, Z)$

X	Y	Z

And suppose I specify $X \to Y$ and also $Y \to Z$ and I stop there.

The reln designer specifies FDs, but there are often more of them that derive from the given ones.

How do you determine the rest of the FDs, so you can use them in your defn?

Armstrong's Axioms (reflexivity)

① If $Y \subseteq X$, then $X \to Y$.

we are allowing these symbols to be sets of attributes

e.g. $\{name\} \subseteq \{class\ year, name\}$

so

$\{class\ year, name\} \to \{name\}$

② If $X \to Y$, then $XZ \to YZ$ for any Z (augmentation)

e.g.

if year → dean

then {year, address} → {dean, address}

③ If $X \to Y$, and $Y \to Z$,

then $X \to Z$ (transitivity)

<u>Closure</u> of a set of <u>FDs</u>.

↳ all the FDs that can be derived from the initial set (and the initial set also.)

<u>Example</u>: $X \to Y$, $Y \to Z$. (suppose one other attr W)

Find the closure: $X \to Y$ $\quad$ $WX \to WZ$

$Y \to Z$

$X \to Z$ $\quad$ augmentation

(augmentation) $WX \to WY$ $\quad$ etc etc etc

Notation: F = set of c FDs

F^+ = closure of F

Normal forms are coming

History / naming is a mess.

The most useful normal form came later.

Codd said "let there be normal forms"
- he had an idea as to what a good reln would look like. The defn was complicated. So he broke it into 3 pieces.

1 NF

2 NF = 1 NF + ...

3 NF = 2 NF + ...

In 1982 Carlo Zaniolo (the person) discovered a much simpler defn for 3NF.

- we will use that.
  - this means 1NF / 2NF are useless today, because Carlo's defn is easier and better.

Mid 70-s, folks discovered a stronger version of 3NF that's usually more desirable w/ some tradeoffs

- Boyce-Codd Normal Form (BCNF)